Assignment No.2

A college is digitizing student records to simplify academic administration. Design a Student class that stores
basic student information and displays the details whenever requested by the administrative staff.

Solution

```
#include <iostream>
#include <string>

using namespace std;

// Class representing a Student record
class Student {
private:
    int rollNumber;
    string name;
    string course;
    int admissionYear;

public:
    // Function to take input for student details
    void inputDetails() {
        cout << "Enter Roll Number: ";
        cin >> rollNumber;
        cin.ignore(); // Clear newline left in the buffer

        cout << "Enter Name: ";
        getline(cin, name);

        cout << "Enter Course: ";
        getline(cin, course);

        cout << "Enter Admission Year: ";
        cin >> admissionYear;
        cout << "-----------------------------------\n";
    }

    // Function to display student details
    void displayDetails() const {
        cout << "Roll Number   : " << rollNumber << "\n";
        cout << "Name          : " << name << "\n";
        cout << "Course        : " << course << "\n";
        cout << "Admission Year: " << admissionYear << "\n";
```

```
        cout << "-----------------------------------\n";
    }

    // Getter function for rollNumber
    int getRollNumber() const {
        return rollNumber;
    }
};

int main() {
    const int MAX_STUDENTS = 100;
    Student database[MAX_STUDENTS];
    int currentCount = 0;
    int choice;

    do {
        cout << "\n===== Student Management System =====\n";
        cout << "1. Add Student Record\n";
        cout << "2. Display All Student Records\n";
        cout << "3. Search Student by Roll Number\n";
        cout << "4. Exit\n";
        cout << "Enter your choice (1-4): ";
        cin >> choice;
        cout << "\n";

        switch (choice) {
            // Add a new student record
            case 1: {
                if (currentCount < MAX_STUDENTS) {
                    cout << "--- Enter Student Details ---\n";
                    database[currentCount].inputDetails();
                    currentCount++;
                    cout << "Student record added successfully!\n";
                } else {
                    cout << "Database is full! Cannot add more students.\n";
                }
                break;
            }

            // Display all student records
            case 2: {
                if (currentCount == 0) {
                    cout << "No student records available to display.\n";
                } else {
```

```
            cout << "--- List of All Students ---\n";
            for (int i = 0; i < currentCount; i++) {
                database[i].displayDetails();
            }
        }
        break;
    }

    // Search for a student using Roll Number
    case 3: {
        if (currentCount == 0) {
            cout << "No student records available to search.\n";
            break;
        }

        int searchRoll;
        bool found = false;

        cout << "Enter Roll Number to search: ";
        cin >> searchRoll;
        cout << "\n";

        // Search through the student records
        for (int i = 0; i < currentCount; i++) {
            if (database[i].getRollNumber() == searchRoll) {
                cout << "--- Record Found ---\n";
                database[i].displayDetails();
                found = true;
                break; // Stop searching once found
            }
        }

        // Display message if student is not found
        if (!found) {
            cout << "Student with Roll Number " << searchRoll << " not found.\n";
        }
        break;
    }

    // Exit the program
    case 4:
        cout << "Exiting system. Goodbye!\n";
        break;
```

```
            // Invalid choice
            default:
                cout << "Invalid choice! Please choose between 1 and 4.\n";
        }

    } while (choice != 4);

    return 0;
}
```

